

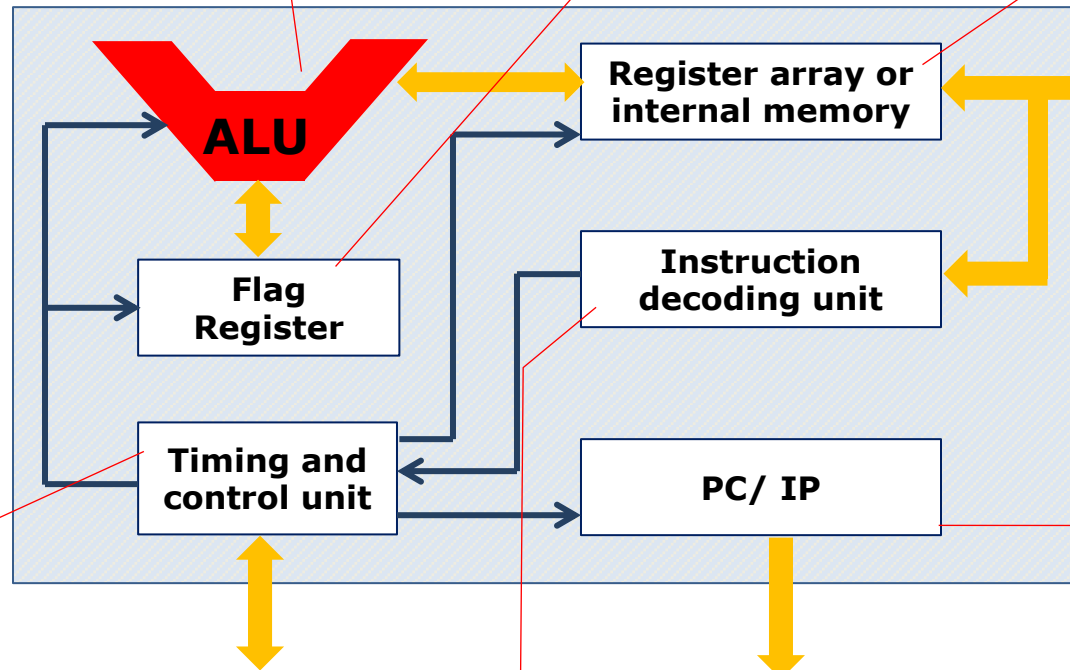
Internal Architecture of 8086 Microprocessor

Functional blocks

Computational Unit; performs arithmetic and logic operations

The different conditions that come from the result of an operation are saved as special bits called flags inside the flag register.

Internal storage of data



Generates the address of the instructions to be fetched from the memory and send through address bus to the memory

Generates control signals for internal and external operations of the microprocessor

Decodes instructions; sends information to the timing and control unit

Overview of 8086 Microprocessor

First 16-bit processor released by INTEL in the year 1978

Originally HMOS, now manufactured using HMOS III technique

Approximately 29,000 transistors, 40 pin DIP, 5V supply

Does not have internal clock; external asymmetric clock source with 33% duty cycle

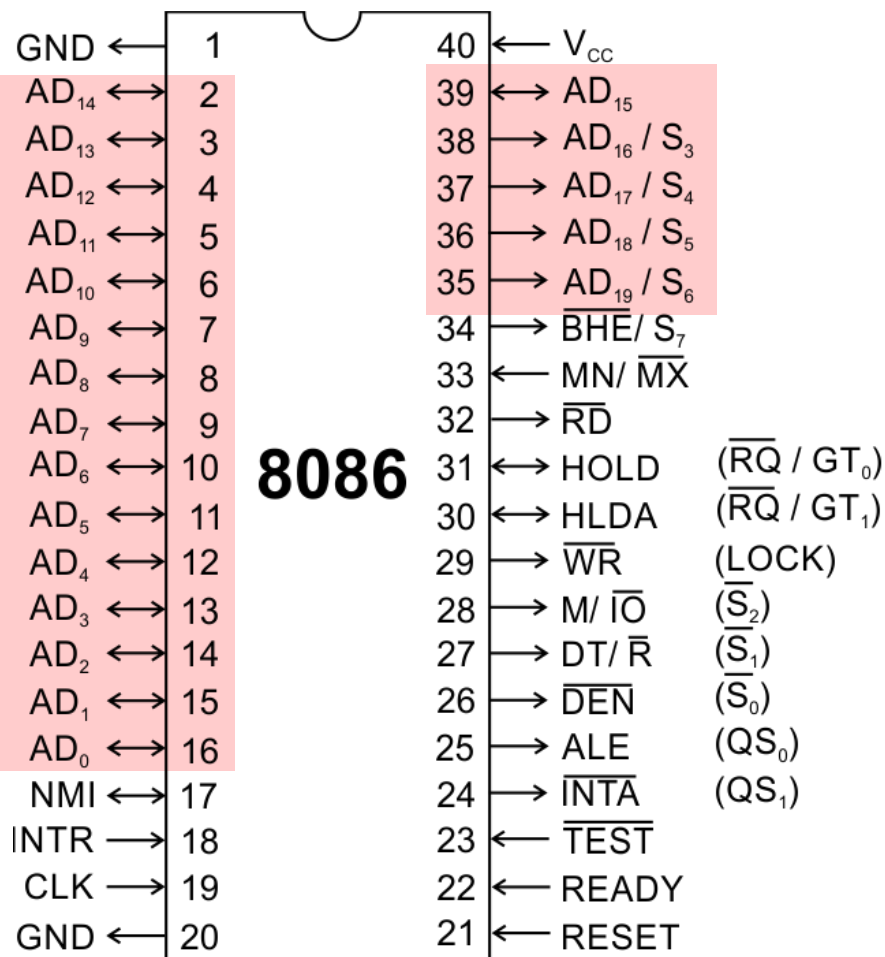
20-bit address to access memory $\Rightarrow$ can address up to $2^{20} = 1$ megabytes of memory space.

Addressable memory space is organized into two banks of 512 kb each; **Even (or lower) bank and **Odd (or higher) bank**. Address line A_0 is used to select even bank and control signal $\overline{BHE}$ is used to access odd bank**

Uses a separate 16 bit address for I/O mapped devices $\Rightarrow$ can generate $2^{16} = 64$ k addresses.

Operates in two modes: **minimum mode and **maximum mode**, decided by the signal at $\overline{MN}$ and $\overline{MX}$ pins.**

Pins and signals



AD₀-AD₁₅ (Bidirectional)

Address/Data bus

The **AD₀-AD₁₅** lines in the 8086 are bidirectional. They work as a multiplexed bus, which means the same lines are used for two purposes:

1. Address bus (to carry memory/IO addresses).
2. Data bus (to carry actual data).

When these lines carry address information, we write them as A₀-A₁₅.

Example: A₀-A₁₅ means they are being used as the lower 16-bit address bus.

When these lines carry data information, we write them as D₀-D₁₅.

Example: D₀-D₁₅ means they are being used as the 16-bit data bus.

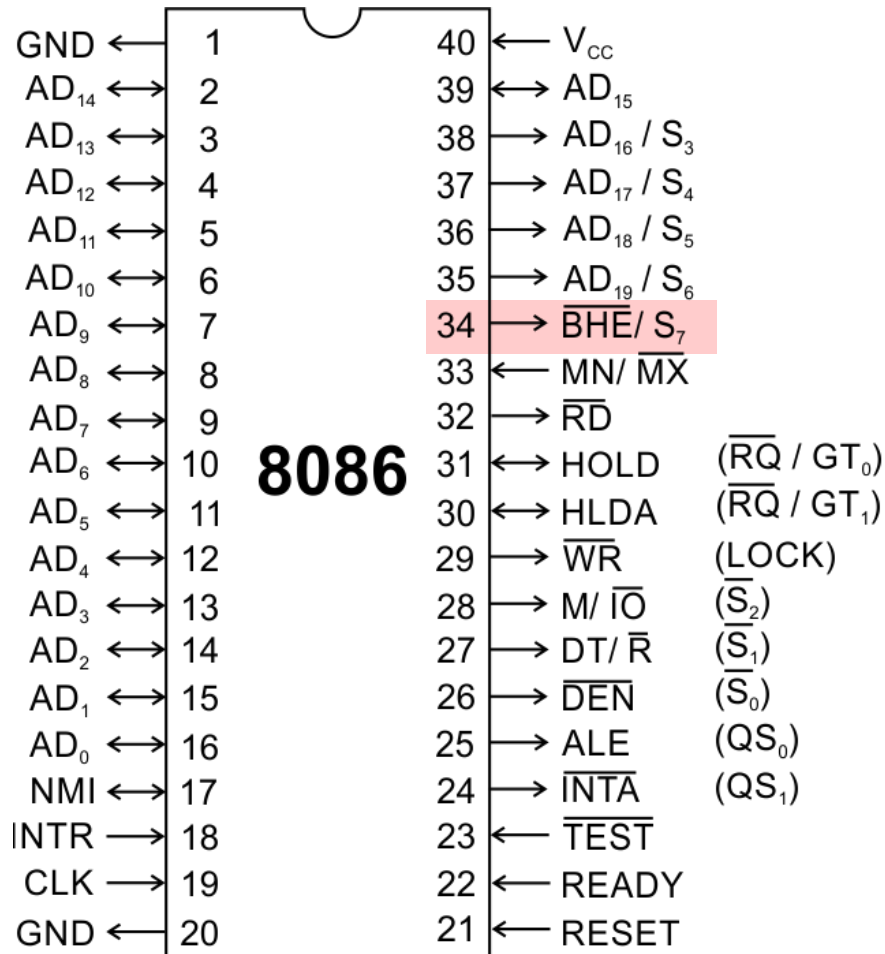
Sometimes, if only part of the bus is used:

D₀-D₇ = lower 8 bits of data.

D₈-D₁₅ = higher 8 bits of data.

A₁₆/S₃, A₁₇/S₄, A₁₈/S₅, A₁₉/S₆

High order address bus. These are multiplexed with status signals



BHE (Active Low)/ S_7 (Output)

Bus High Enable/Status

It is used to enable data onto the most significant half of data bus, D_8-D_{15} . 8-bit device connected to upper half of the data bus use BHE (Active Low) signal. It is multiplexed with status signal S_7 .

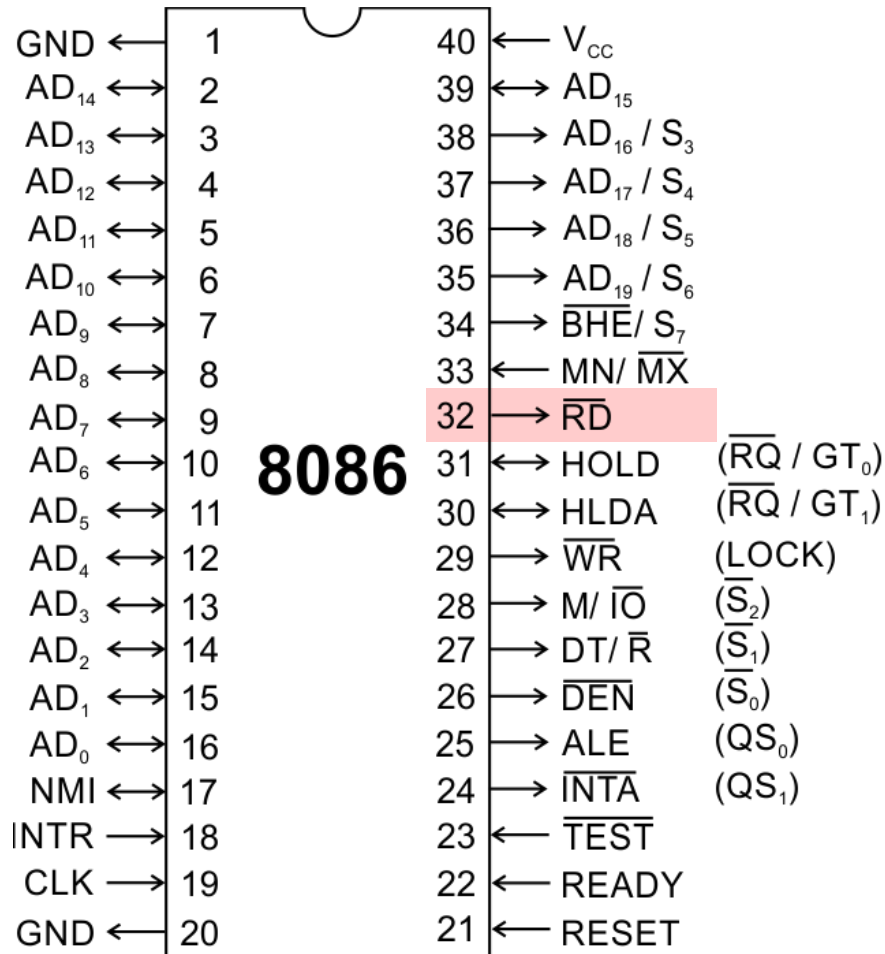
MN/ MX

MINIMUM / MAXIMUM

This pin signal indicates what mode the processor is to operate in.

RD (Read) (Active Low)

The signal is used for read operation.
It is an output signal.
It is active when low.

**TEST**

$\overline{TEST}$ input is tested by the 'WAIT' instruction.

8086 will enter a wait state after execution of the WAIT instruction and will resume execution only when the $\overline{TEST}$ is made low by an active hardware.

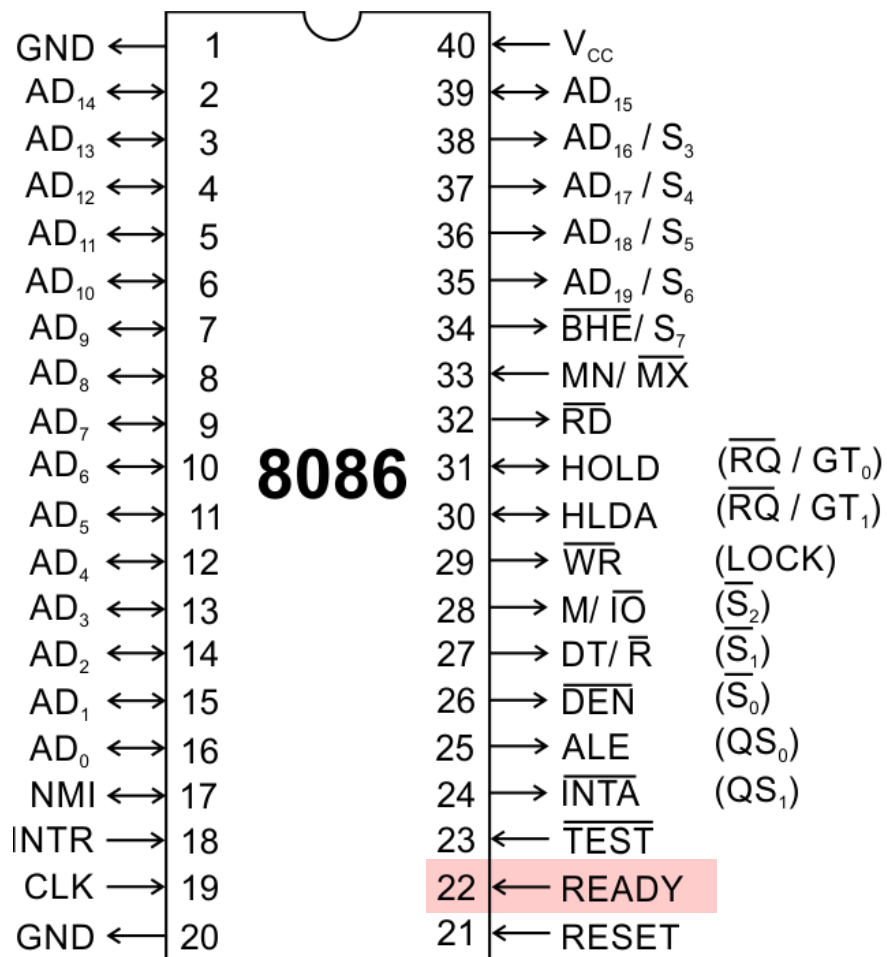
This is used to synchronize an external activity to the processor internal operation.

READY

This is the acknowledgement from the slow device or memory that they have completed the data transfer.

The **READY** pin is used by slow memory or I/O devices to tell the CPU whether they are ready for data transfer.

The signal is active high.



RESET (Input)

Causes the processor to immediately terminate its present activity.

The signal must be active HIGH for at least four clock cycles.

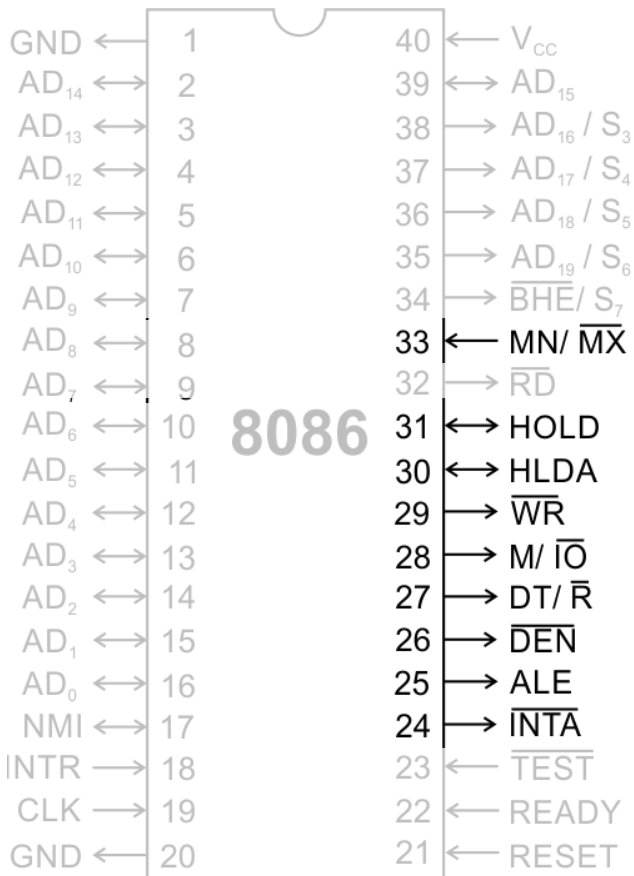
CLK

The clock input gives the basic timing for the processor to work and to control the bus. It is an uneven square wave signal that stays high for 33% of the time and low for the rest.

INTR Interrupt Request

This is a triggered input. It is checked at the end of each instruction to see if there is an interrupt request. If a request is present, the processor starts the interrupt acknowledge cycle.

This signal is active high and is synchronized inside the processor.



The 8086 microprocessor can work in two modes of operations : **Minimum mode** and **Maximum mode**.

In the minimum mode of operation the microprocessor do not associate with any co-processors and can not be used for multiprocessor systems.

In the maximum mode the 8086 can work in multi-processor or co-processor configuration.

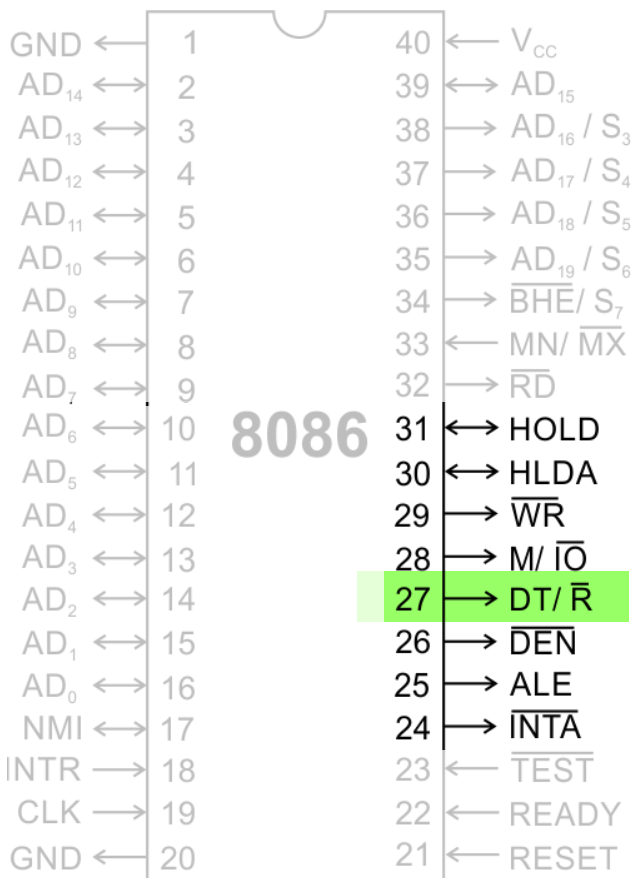
Minimum or maximum mode operations are decided by the pin MN/ MX(Active low).

When this pin is high 8086 operates in minimum mode otherwise it operates in Maximum mode.

Pins 24 -31

For minimum mode operation, the $\overline{MN}/\overline{MX}$ is connected to VCC (logic high)

8086 itself generates all the bus control signals

**DT/ $\overline{R}$**

(**Data Transmit/ Receive**) It is a control signal used to manage direction of data flow between the CPU and peripherals through transceivers.

 $\overline{DEN}$

(**Data Enable**) It enables external devices to read data from the 8086 by indicating that valid data is present on the data bus.

ALE

(**Address Latch Enable**) Used to demultiplex the address and data lines using external latches

M/ $\overline{IO}$

Used to differentiate memory access and I/O access. For memory reference instructions, it is **high**. For IN and OUT instructions, it is **low**.

 $\overline{WR}$

Write control signal; asserted **low** Whenever processor writes data to memory or I/O port

 $\overline{INTA}$

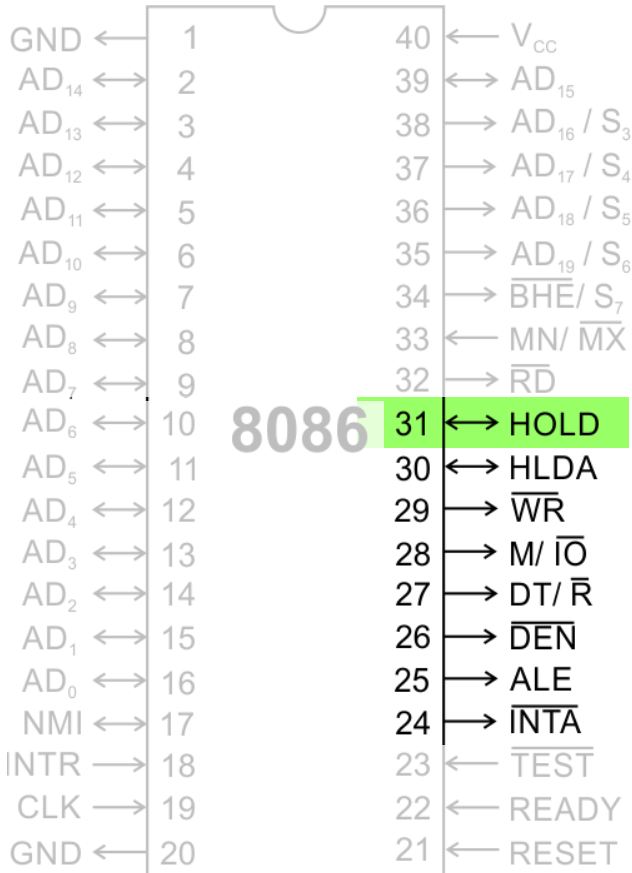
(**Interrupt Acknowledge**)

When an external device requests an interrupt by activating the INTR (Interrupt Request) pin, the 8086 responds if interrupts are enabled.

Pins 24 -31

For minimum mode operation, the $\overline{MN}/\overline{MX}$ is tied to VCC (logic high)

8086 itself generates all the bus control signals

**HOLD**

The HOLD pin indicates that an external device requests control of the 8086's buses, and the processor will release them after completing the current operation.

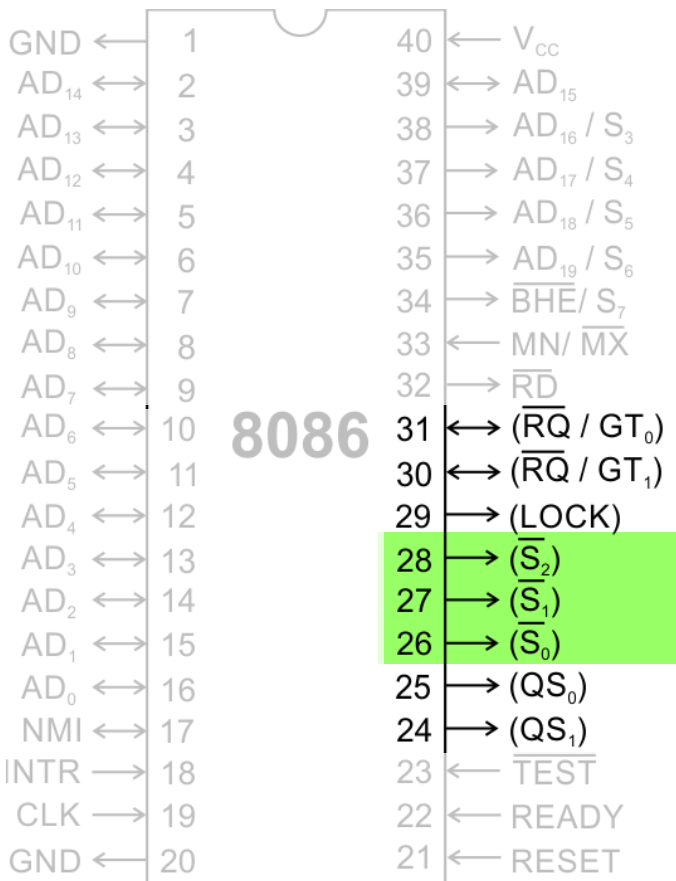
It is usually used by the DMA controller to take control of the bus.

HLDA

(Hold Acknowledge) The HLDA pin goes high to acknowledge that the 8086 has released the buses and an external device can take control.

During maximum mode operation, the $\overline{\text{MN}}/\overline{\text{MX}}$ is grounded (logic low)

Pins 24 -31 are reassigned



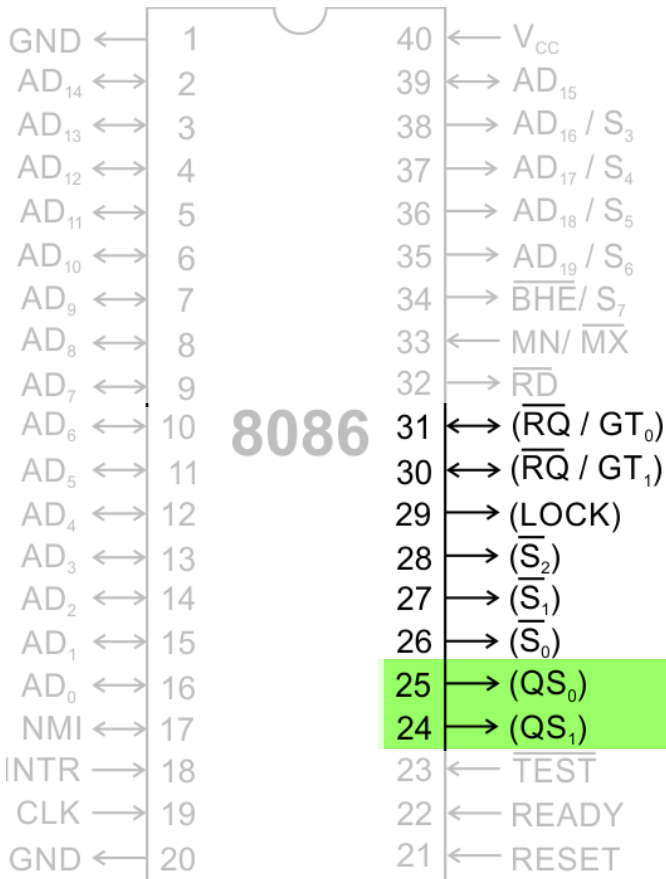
$\overline{\text{S}}_0, \overline{\text{S}}_1, \overline{\text{S}}_2$

Status signals; used by the 8086 bus controller to generate bus timing and control signals. These are decoded as shown.

S_2	S_1	S_0	Function (Bus Cycle Type)
0	0	0	Interrupt Acknowledge
0	0	1	Read I/O Port
0	1	0	Write I/O Port
0	1	1	Halt
1	0	0	Instruction Fetch
1	0	1	Read Memory
1	1	0	Write Memory
1	1	1	Passive (Invalid/Idle State)

During maximum mode operation, the $\overline{MN}/\overline{MX}$ is grounded (logic low)

Pins 24 -31 are reassigned



QS_0, QS_1 (Queue Status) The processor provides the status of queue in these lines. The queue status can be used by external device to track the internal status of the queue in 8086.

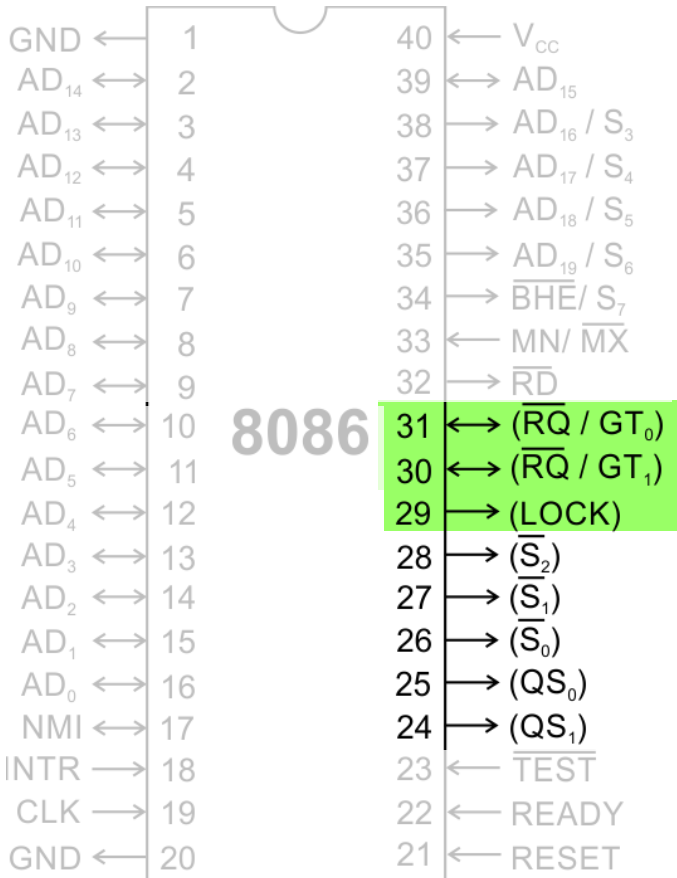
The output on QS_0 and QS_1 can be interpreted as shown in the table.

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QS_0	QS_1	Status
0	0	No operation
0	1	First byte of opcode from queue
1	0	Queue empty
1	1	Subsequent byte of opcode from queue

During maximum mode operation, the $\overline{MN}/\overline{MX}$ is grounded (logic low)

Pins 24 -31 are reassigned



**$\overline{RQ}/GT_0$
 $\overline{RQ}/GT_1$**

(**Bus Request/ Bus Grant**) In the 8086 microprocessor, the BR (Bus Request) pin is used by an external device to ask for control of the buses, and the BG (Bus Grant) pin tells the device that the 8086 has given the buses, so the device can use them safely.

LOCK

In the 8086 microprocessor, the LOCK pin is used to make sure the processor has exclusive control of the system bus during critical instructions, preventing other devices or processors from accessing the bus until the instruction is finished.